F16GE104-CB

400A
4 POLE

G
3~

F16-GE-104
415V
250kVA

F16-SR-002
415V FLOTATION SWITCHROOM

SWR EARTH BAR

FEEDER FROM F16-SW-004
2000-F16-SLD-EL-10201
TO F16-SW-004
2000-F16-SLD-EL-10201

F16-SW-104
FLOTATION 415V ESSENTIAL SWITCHBOARD

EARTH BAR

NEUTRAL BAR
NOTE 1

AFD

125A
160A

SD
NOTE 3

2A
63A

2A
63A

PFR

2A
63A

MANUAL
MCCB CHANGEOVER
SWITCH 3 POLE 400A

415V 3NE~ 50Hz 400A 65kA for 1sec

00F-05
INC

01F-04

63A

2A
32A

70A

63A
63A

01F-09

250A

2A
32A

90A

1F-01

250A

2A
32A

90A

01F-13

250A

2A
32A

90A

ISL

30kW

F16UPS002-ISL
415VAC UPS
MAINTENANCE
BYPASS SWITCH
2000-F16-SCM-EL-10215

F16-DB-202
FLOTATION
415V SWITCHROOM
415VAC UPS DB
2000-F16-TLD-EL-10056

UPS

30kVA

F16-UPS-002
FLOTATION
415V SWITCHROOM
415VAC UPS
2000-F16-TLD-EL-10056

JB F17DB301-JB

JB F18DB301-JB

JB F16DB301-JB

69kW

F17-DB-301
FLOTATION
TAILINGS THICKENER AREA
L&SP DB
2000-F17-SCM-EL-10201

69kW

F18-DB-301
CLEANER
FLOTATION AREA
L&SP DB
2000-F18-SCM-EL-10201

69kW

F16-DB-301
FLOTATION AREA
L&SP DB
2000-F16-SCM-EL-10205

F16-SR-001
11KV FLOTATION
SWITCHROOM

15kW

F16-DB-201
FLOTATION
11KV SWITCHROOM
415VAC UPS DB
2000-F16-TLD-EL-10057

NOTES

1. EARTH-NEUTRAL LINK INSTALLED IN UPSTREAM F16-SW-004 INCOMER TIER. SIGNAGE SHALL BE PLACED ON F16-SW-104 SWITCHBOARD AND ON GENERATOR READING "M.E.N. LINK LOCATED IN F16-SW-004 INCOMER TIER"
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
4. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
5. CIRCUIT BREAKER TO BE SUBSTITUTED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.

LEGEND

PFR	PHASE FAILURE RELAY
/ >	EARTH LEAKAGE RELAY NOTE 4, 5
xxxA / xxxA	ST TRIP SET POINT / CB SIZE
AFD	ARC FLASH DETECTION
ECR	EARTH CONTINUITY RELAY

2000-F16-SLD-EL-10251_Rev2

F16-SW-104 415V essential switchboard - single line diagram

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\235171 KCGM GROWTH PROJECT\03 DELIVERABLES\ELV\F16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SLD-EL-10251.DWG 03SEP26 3:07PM ROBINLAW